CSE 3400/CSE 5850
Introduction to Cryptography and Cybersecurity/
Introduction to Cybersecurity
School of Computing

Syllabus -- Fall 2026

Course Info

Title. Introduction to Cryptography and Cybersecurity (or Introduction to Cybersecurity)
Credits. 3.00 credits
Format. In person
Meeting time and location. Section 1: Tue/Thu 9:30 - 10:45 am, ITE 127
Section 2: Tue/Thu 2:00 - 3:15 pm, MCHU 305

Some classes could be synchronous virtual or asynchronous (where a video of the class will be posted on HuskyCT before the class starts). Synchronous virtual/Asynchronous classes will be announced in advance.

Course Description and Objectives

Introduction to applied cryptography and cybersecurity. The course covers basic elements of symmetric-key and public-key cryptography, including encryption, authentication, signatures and key agreement, as well as integrity (hash functions, blockchains and more), pseudorandomness, shared key protocols, and the Transport Layer Protocol (TLS). Discuss principles of cybersecurity and ethics.

The course will allow students to develop the security-thinking approach or mindset; define security goals, violations, attackers capabilities and their goals, design secure schemes and protocols, and then argue (somehow) formally about the security of these schemes and protocols. This is in addition to studying many practical vulnerabilities, attacks and defenses.

Instructor and Contact Info

Ghada Almashaqbeh
Email: ghada@uconn.edu

Office Hours. Every Tuesday 1 - 2 pm at my office ITE 255 (any changes will be announced on HuskyCT) , or by appointment (if you cannot make it to the office hour, please email me to arrange another time).

TAs

Maryam Rezapour, email: maryam.rezapour@uconn.edu
Office Hours. Every Wednesday 4 - 5:30 pm in ITE 227 and ITE 214 (first check ITE 227, if Maryam is not there then go to ITE 214).

Colin Finkbeiner, email: colin.finkbeiner@uconn.edu
Office Hours. Every Thursday 11:30 am - 1 pm in ITE 214.

Communication

The lecture slides will be posted on the course website (usually the night before the class): https://ghadaalmashaqbeh.github.io/courses/cse3400-f2026/ . Syllabus, announcements, problem sets, solutions, homework submission, etc., will be posted on/via HuskyCT. We will also have a discord server for the course for questions/discussions. This will be set up by the TAs and announced separately. Please use it for general questions (problem sets, course material, logistics, etc.) but not to answer the homeworks/exams or ask questions that expose the answers indirectly.

I will be answering questions on discord on a daily basis (usually once a day), the TAs will be following these questions as well at a higher frequency. For emails, I will answer in 24 - 48 hours once I receive your email. If for some reason you do not hear back from me within this timeframe, please feel free to send me a reminder. I will not answer emails over the weekend (Friday 5 pm until Monday 8 am).

Regarding points reviews/disputes; any requests to review graded homeworks/exams/etc. should be discussed with the instructor not the TAs.

Textbook

Applied introduction to Cryptography and Cybersecurity*, by Amir Herzberg, 2026.*

- Published by World Scientific Publishing (https://www.worldscientific.com/). Available from the publisher (also as ebook) at https://www.worldscientific.com/worldscibooks/10.1142/14014#t=aboutBook or from Amazon
- It is enough for the course to use an early draft of the book (I will use it when citing sections/chapters throughout the course), download a copy using this link (reach out to the instructor/TAs if you have any problems with downloading the draft) https://drive.google.com/file/d/19WZi47ZRGs6SzCh6JG5CAti8mEQjAVW_/view

Students are encouraged to read lecture notes/slides before and after lectures, and to solve exercises and examples (found in the textbook), preferably trying to do so without viewing the solutions and only then checking against the solutions (where available).

Note: the draft URL (https://sites.google.com/site/amirherzberg/crypto-cyber-book?authuser=0) also contains the textbook slides, but we will not use them. The slides used in class are an adaptation of these on the textbook website (I add extra information and exercise problems/examples, remove uncovered material from the slides, etc.). Make sure to fetch the class slides from the course website https://ghadaalmashaqbeh.github.io/courses/cse3400-f2026/ .

Grading and Course Work

CSE 3400 students

Quizzes	6%
Homeworks	34%
Midterm test I	30%
Midterm test II	30%

CSE 5850 students

Quizzes	6%
Homeworks	30%
Term Project	10%
Midterm test I	27%
Midterm test II	27%

Homeworks will consist of a small number of problems covering the material taken in class. We will have 3 homeworks during the semester. All submissions must be PDF format, the use of Latex is highly encouraged (you can use an online software for that like overleaf.com). You can use other word processors (like MS word) but the final submission has to be converted to a PDF.

Submission and late policy. Each student will get 6 (free) late days, if a student uses all their free days, then grade deduction will be applied: a 15% deduction of homework score per day. **For each homework, up to 3 delay days at maximum can be used**, thus for any homework a student can use up to 3 late days (free if they did not use all of their 6 late days, or grade-deducted ones if they have already used all 6 free late days). **After the deadline and late period elapses, no submissions will be accepted!** TAs will track late days used across homeworks for all students. Assignments will be graded no more than two weeks after they are due and the key solution will be posted.

Collaboration. Homeworks ***must be done and submitted individually.*** However, students can discuss high level ideas with each other given that they write their solutions individually and list the names of the students with whom they collaborated in the submission. Copied solutions are considered cheating, and same for using AI tools to solve the homework. You can collaborate with another two students at maximum (i.e., a total of 3 students at max can discuss with each other). I encourage you to solve the problems on your own first and then resort to group discussion for further understanding and brainstorming.

Midterm tests. There will be two midterm tests; each is two hours and **in-person**. Tests are done individually, no collaboration between students is allowed, and questions about the tests are restricted to clarifying the problems but not about the correctness of the solution (or getting hints on how to solve a problem). Tests are open book/slides/notes but all these must be hard copies; no electronic devices of any type are allowed. The use of the Internet/other textbooks/AI tools or asking others (from inside or outside the class) are strictly prohibited. Attempting any of these will be considered cheating.

Test	**Date and Time**	**Location**
Midterm test I	Thursday, 10/15/2026 @ 5 pm - 7 pm	ARJ 143
Midterm test II	Thursday, 12/10/2026 @ 5 pm - 7 pm	ARJ 143

Quizzes. There will be 2-4 **in-class/in-person** quizzes (based on course progress); quiz dates and covered material will be announced in advance.

Term project (CSE 5850). Groups of exactly 2 students. Choose a cryptography or cybersecurity topic (theory or applied) and prepare a presentation to explain this topic for the class. More details about the project will be posted on HuskyCT throughout the semester. The project will have these milestones:

- Proposal (containing the group names and the topic title that will be covered) 1%, submission deadline is 10/8/2026 by midnight.
- Presentation and discussion 9%, presenting/discussing the chosen topic to demonstrate understanding and critical thinking. Presentations/discussions will take place during the last week of classes in the semester (the schedule will be announced later).

Grading Scale. Letter grades will be assigned based on the following grade scale.

Grade	Letter Grade
[90, 100]	A
[86, 90)	A-
[82, 86)	B+
[79, 82)	B
[76, 79)	B-
[73, 76)	C+
[70, 73)	C
[67, 70)	C-
[64, 67)	D+
[60, 64)	D
[55, 60)	D-
[0, 55)	F

Tentative Course Schedule

Week of	**Topic**
8/31/2026	Introduction, encryption
9/7/2026	Encryption
9/14/2026	Encryption and pseudo-randomness
9/21/2026	Pseudorandom functions and block ciphers
9/28/2026	Authentication
10/5/2026	Hashing
10/12/2026	Hashing and Blockchains ***Midterm Test I (no classes on the test date 10/15/2026, we will meet for the test instead)***
10/19/2026	Hashing and Shared key protocols
10/26/2026	Shared key protocols
11/2/2026	Key Exchange and Recovery
11/9/2026	Public Key Cryptosystems
11/16/2026	Public Key Cryptosystems
11/23/2026	***Thanksgiving Recess - no classes!***
11/30/2026	PKI
12/7/2026	CSE 5850 project presentations during the classes on 12/8/2026. ***Midterm test II (no classes on the test date 12/10/2026, we will meet for the test instead).***

Policies

Academic honesty. This course expects all students to act in accordance with the Guidelines for Academic Integrity at the University of Connecticut. Additionally, consult UConn's guidelines for academic integrity. The collaboration policy described above is designed to allow students the resources to succeed while ensuring they learn and master the material. If you are unsure if something is acceptable according to the collaboration policy, talk to me!

Violations of this policy will be considered violations of the academic integrity policy and will be reported to the Academic Integrity Hearing Board. Consequences may include (but are not limited to) failure of the class. Example violations include: not reporting collaborators, jointly writing solutions, copying or plagiarizing solutions and projects from other sources.

Student conduct code. Students are expected to conduct themselves in accordance with UConn's student conduct code (https://community.uconn.edu/the-student-code/).

Copyright. My lectures, notes, handouts, and displays are protected by state common law and federal copyright law. Students may take notes. In addition, students will be consulted before using their solutions either with or without their name.

Students with Disabilities. The University of Connecticut is committed to protecting the rights of individuals with disabilities and assuring that the learning environment is accessible. If you are a student with approved academic accommodations through the Center for Students with Disabilities (CSD), please let me know immediately so we can discuss implementation. If you anticipate or experience any physical or academic barriers based on disability or pregnancy, you should contact the CSD to request accommodations at csd@uconn.edu or (860) 486-2020. Information about requesting accommodations is available on the CSD website at http://csd.uconn.edu/

Resources for Students Experiencing Distress. The University of Connecticut is committed to supporting students in their mental health, their psychological and social well-being, and their connection to their academic experience and overall wellness. The university believes that academic, personal, and professional development can flourish only when each member of our community is assured equitable access to mental health services. The university aims to make access to mental health attainable while fostering a community reflecting equity and diversity and understands that good mental health may lead to personal and professional growth, greater self-awareness, increased social engagement, enhanced academic success, and campus and community involvement.
Students who feel they may benefit from speaking with a mental health professional can find support and resources through the Student Health and Wellness-Mental Health (SHaW-MH) office. Through SHaW-MH, students can make an appointment with a mental health professional and engage in confidential conversations or seek recommendations or referrals for any mental health or psychological concern.

Accommodations for Illness or Extended Absences. If illness prevents you from attending class, it is your responsibility to notify your instructor as soon as possible with proper documentation. You do not need to disclose the nature of your illness, however, you will need to work with your instructor to determine how you will complete coursework during your absence.
If life circumstances are affecting your ability to focus on courses and your UConn experience, students can email the Dean of Students at dos@uconn.edu to request support. Regional campus students should email the Student Services staff at their home campus to request support and faculty notification.

AI policy for students. All work submitted in this course must comply with UConn's Academic, Scholarly, and Professional Integrity and Misconduct Policy and, for undergraduate courses, Appendix A of The

Student Code. Students are responsible for submitting their own work, acknowledging the contributions of others, and avoiding unauthorized assistance.

You may not use generative AI tools to generate, revise, summarize, analyze, solve, or otherwise contribute to work submitted in this course unless an assignment explicitly permits their use. The work in this course is designed to enable students to build their own skills and understanding of cryptography and security concepts. Developing such ability depends on having students understand course material and do the assigned work (exams, quizzes, homeworks, etc.) by themselves. Accordingly, students must solve homeworks, exams, quizzes, and prepare term projects by themselves and are responsible for their solutions/answers/submissions.

Assignment-specific directions take precedence over this statement. Permission to use generative AI for one assignment does not constitute permission to use it for other assignments.

Standard spelling and grammar check, accessibility tools, and any tool approved as part of an academic accommodation are not restricted by this statement. Do not enter another person's unpublished work, personal information, or sensitive data into unsupported third-party generative AI tools. When using institutionally supported tools such as Microsoft Copilot, follow university data governance and privacy guidelines. You are responsible for understanding whether the tools you use include generative AI features. If you are unsure whether a tool or feature is permitted, ask the instructor before using it.

Using generative AI without authorization or beyond the permitted limits may constitute academic misconduct under UConn policy. If the instructor believes work submitted by a student includes unauthorized generative AI use, I will talk with the student first about it before deciding whether to refer the matter to Community Standards.

Instructor use of generative AI. The instructor/TAs do not use generative AI to design this course, develop its materials, respond to student work, or communicate with students. If that changes during the semester, we will let students know.